27th International Congress of the Mediterranean Federation for Health and Production of Ruminants (Fe.Me.S.P.Rum.)-2025

Session: Buffalo Chairman: Domenico Vecchio (Italy)

Milk microbiota variability in the quarter and farm bulk milk in the water buffalo milk value chain in Bangladesh

Shuvo Singha*, Md. Nazmul Hoque, Chiara Gini, Paola Cremonesi, Munawar Sultana, Gerrit Koop, Md. Mizanur Rahman, Domenico Vecchio, Ylva Persson, Fabrizio Ceciliani

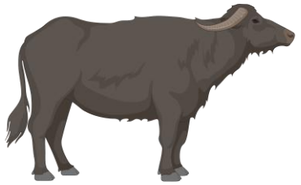
** Postdoc Researcher, Chattogram Veterinary and Animal Sciences University*
Udder Health Bangladesh

Introduction

Features of Water buffalo production

Family: Bovidae

Bubalus bubalis



Characteristics	Water buffalo (Indigenous/ cross-breed)	Water buffalo (Mediterranean)
Age of sexual maturity	**35 - 48 months**	24 months
Reproductive cycle	21 (17-28) days	21 days
Estrus period	1 - 48 hours	1 - 48 hours
Gestation period	300-310 days	310 days
Time between deliveries	15 - 18 months	14 months
Lactation period	**150 days**	270 days
Milk yield per day	**1 - 5 L**	8 - 15 L
Reproductive career	15 - 20 years	15 years

The **sweat glands is very limited** → thermoregulation control

The **teats are longer and with many muscular fibers** → protection from mastitis

Wallowing

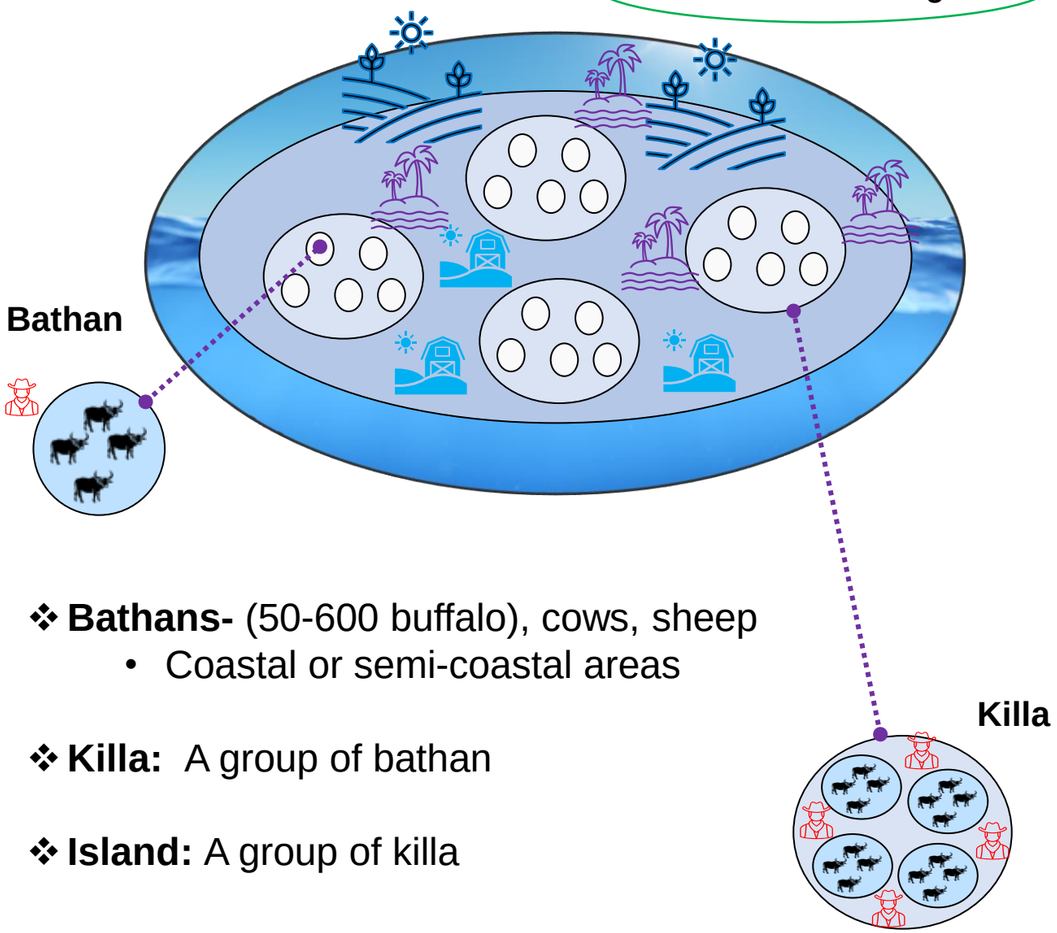
Hamid et al., 2

Introduction

Buffalo rearing systems in Bangladesh

Island

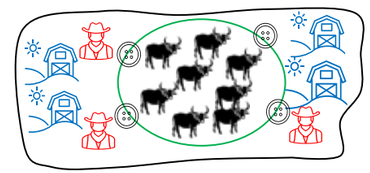
Bathan/ Free-range



- ❖ **Bathans-** (50-600 buffalo), cows, sheep
  - Coastal or semi-coastal areas
- ❖ **Killa:** A group of bathan
- ❖ **Island:** A group of killa

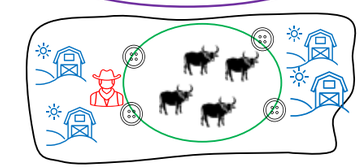
Inland

Semi-bathan

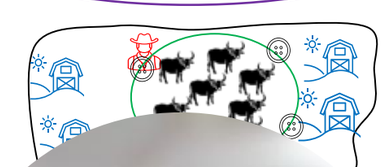


Intensive

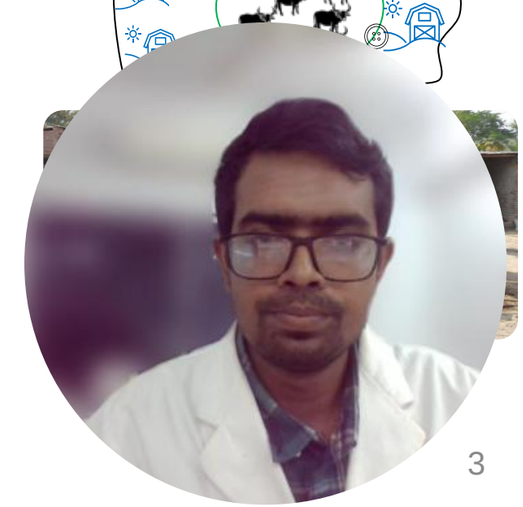
Household



Semi-intensive



- ❖**Semi-bathan-** Buffaloes moved from island to inland
- ❖**Intensive-** Buffalo (20-200 buffalo/ farm)
- ❖**Semi-intensive-** Buffalo (4-15 buffalo/ farm)
- Riverine areas
- ❖ **Household-** Buffalo (1-10 buffalo/ farm)



Introduction

Previous research findings in a PhD in the water buffalo milk chain in Bangladesh

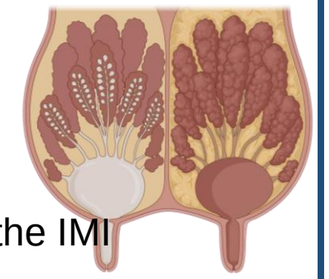
Udder health

Prevalence of SCM

The research gaps related to udder health identified in the systematic review were mostly met

Prevalence was high (animal-52% and quarter-28%)

- ✓ Free-range and semi-intensive system could reduce the SCM
- ✓ Breeding could be useful to control the SCM
  - Teat shape
  - Udder symmetry



Prevalence of IMI

High prevalence of NASM

- ✓ Better management could help reduce the IMI
  - Milking hygiene
  - Cleanliness of hind quarter

BMSCC

Overall BMSCC (217,000 cells/ mL of milk) was satisfactory

But there was a large variation between farms

- ✓ Room for improvement
- ✓ Further studies are needed to identify between farm variations

Milk quality

Buffalo milk value chain

Milk handling operations are diverse in different rearing systems

- ✓ One control measures might not fit to all

BMSCC and bacteriological quality

BMSCC increased in intensive farms

A progressive increase in TBC and TEC over the milk value chain

- ✓ Better management on farms
  - River areas are better than coastal areas
  - Use of tube-well water to clean milk container
- ✓ Better milk handling at middleman
  - Practice of milk transportation with cooling facility
  - Cleaning of milk containers should be improved

Zoonotic pathogens

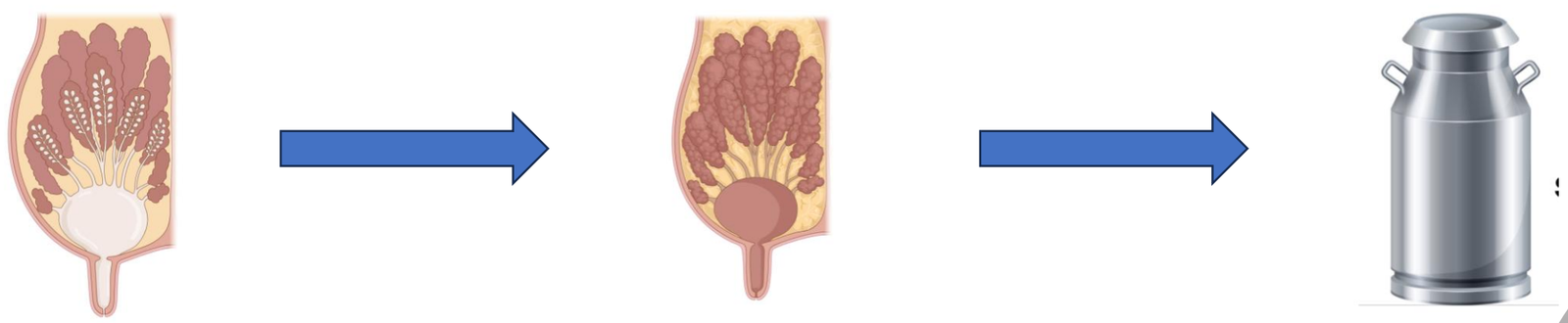
Overall prevalence of most pathogens in milk and milk … *high*

- ✓ Better management on farms
  - District and free-ranging system need furth
- ✓ Better milk handling at milk collection center an
  - Use of same cloth and plastic milk contain

Little is known about the microbiome composition in the buffalo milk chain in Bangladesh

Objectives

This study aimed to examine the bacteriome diversity and composition across stages of the buffalo milk supply chain in Bangladesh

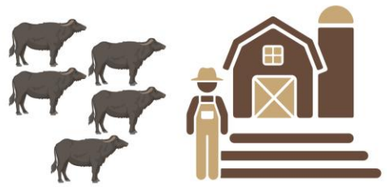


Healthy quarter Subclinical quarter Bulk m

Fe. Me. S. P. Rum.
27° international congress
Bologna, September 4-6, 2025

Materials and Methods

Farm selection



15 farms selected with **1 healthy, 1 subclinical quarter, and 1 bulk milk** samples each farm

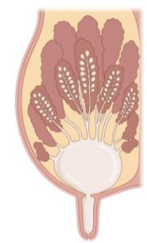
37 free-ranging or semi-free-ranging farms enrolled from **Noakhali, Mirsarai, and Godagari**

Total **45 samples** (**15 HM; 15 SCM; 15 BM**) rocessed for 16s rRNA sequencing

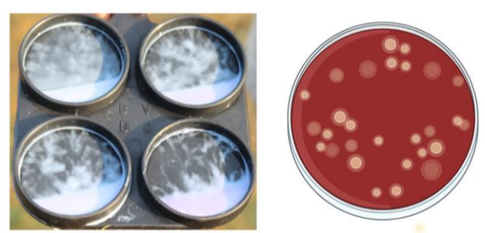
Sampling

Quarter milk sample

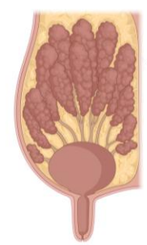
Milk sample is CMT negative and culture negative



Healthy quarter Sample (HM) n = 15



Milk sample is CMT positive and culture positive



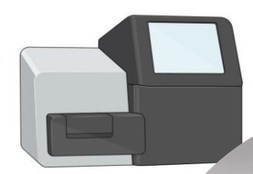
Subclinical quarter Sample (SCM) n = 15

Mixture of milk from all the lactating buffalo in each farm

Bulk milk samples (BM) n = 15

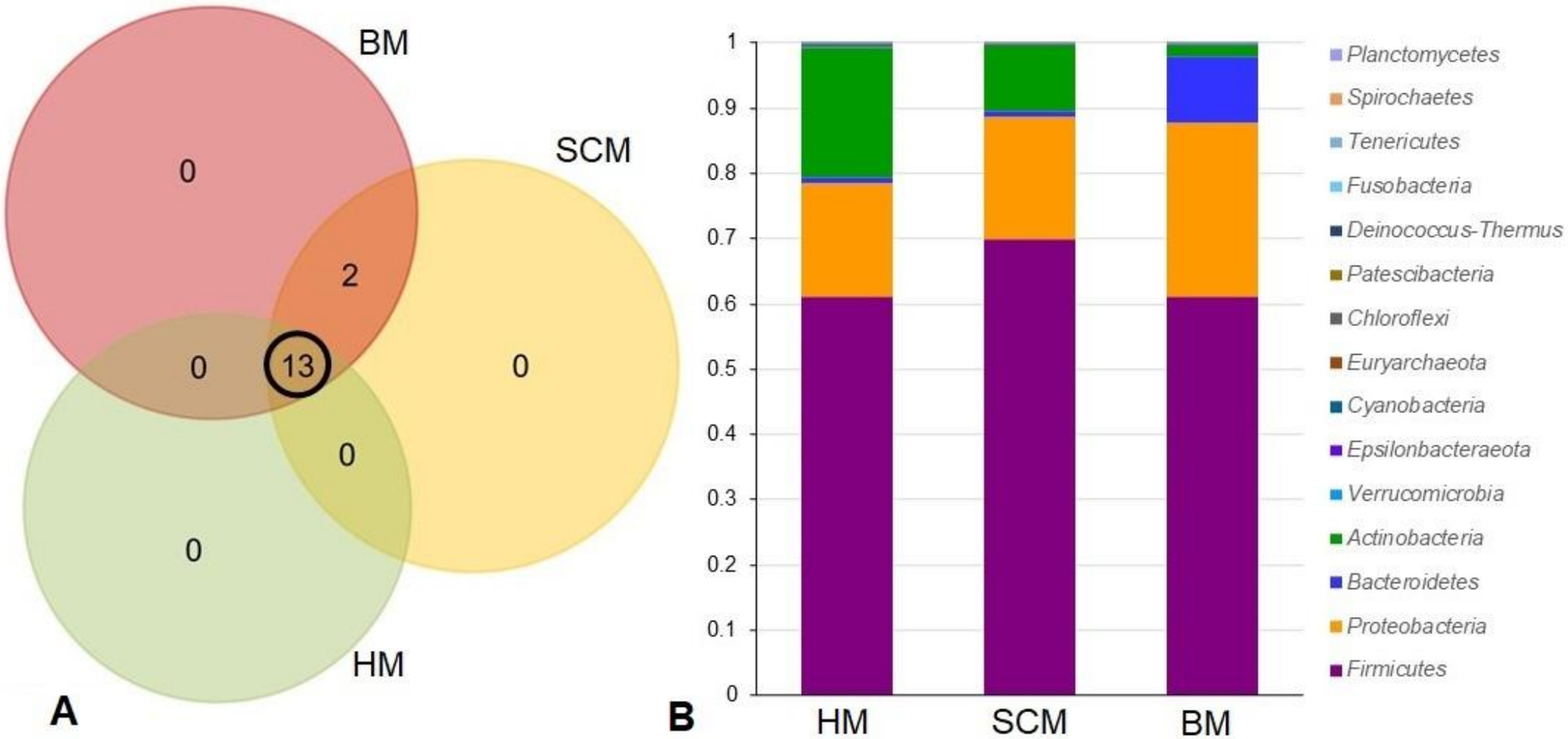
DNA extraction using Maxwell 16 cell DNA Extraction

16s rRNA sequencing using Illumina Miseq



Bioinforma

Results



(A) Out of 15 detected phyla, **13 phyla were found to be shared** in the metagenomes.

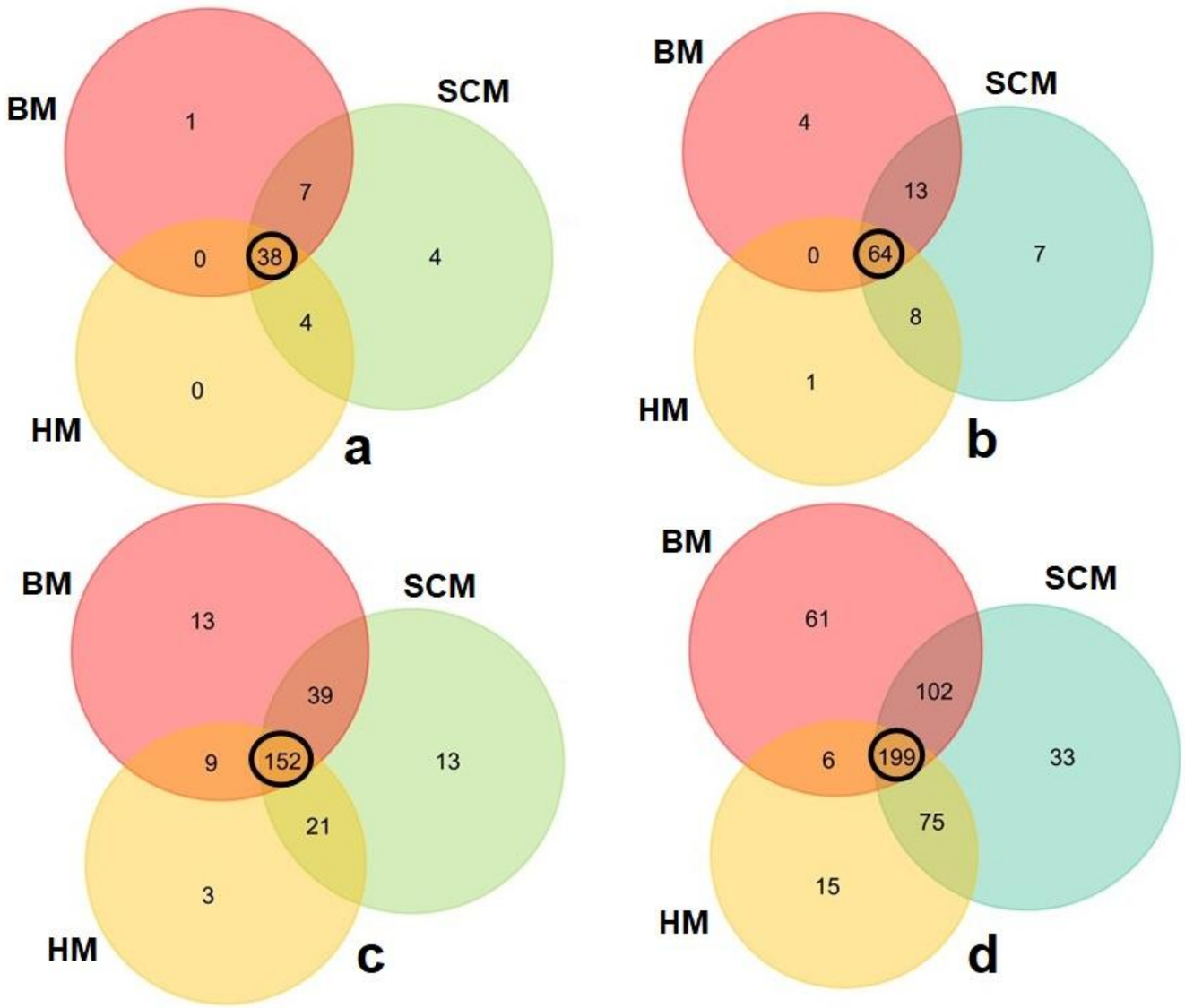
(B) The bar plots representing the relative abundances of detected bacterial phyla in HM, SCM and BM sample groups.

The overall composition and relative **abundance of microbiome always remained higher** in mastitis (SCM) milk samples compared to BM and HM (SCM>BM>HM)

Fig 1. Bacterial phyla detected. The phylum-level taxonomic abundance of microbiomes in SCM (subclinical mastitis milk), BM (bulk milk) and HM (healthy milk) samples.

Fe. Me. S. P. Rum.
27° international congress
Bologna, September 4-6, 2025

Results



Venn diagrams representing the unique and shared bacterial taxa SCM (subclinical mastitis milk), BM (bulk milk) and HM (healthy milk) samples.

(A) 54 orders detected, only **38 shared orders**

(B) 97 bacterial families detected across the sample groups, of which **64 families were shared**

(C) 251 bacterial genera detected, 13, 13, and 3 genera had sole association with SCM, BM and HM subjects, respectively, and **152 genera were shared**

(D) 491 bacterial species, the SCM, BM and HM cases had sole association of 33, 61 and 15 species, respectively, and **199 species were found to b**

Fig 2. Taxonomic composition of microbiomes

Results

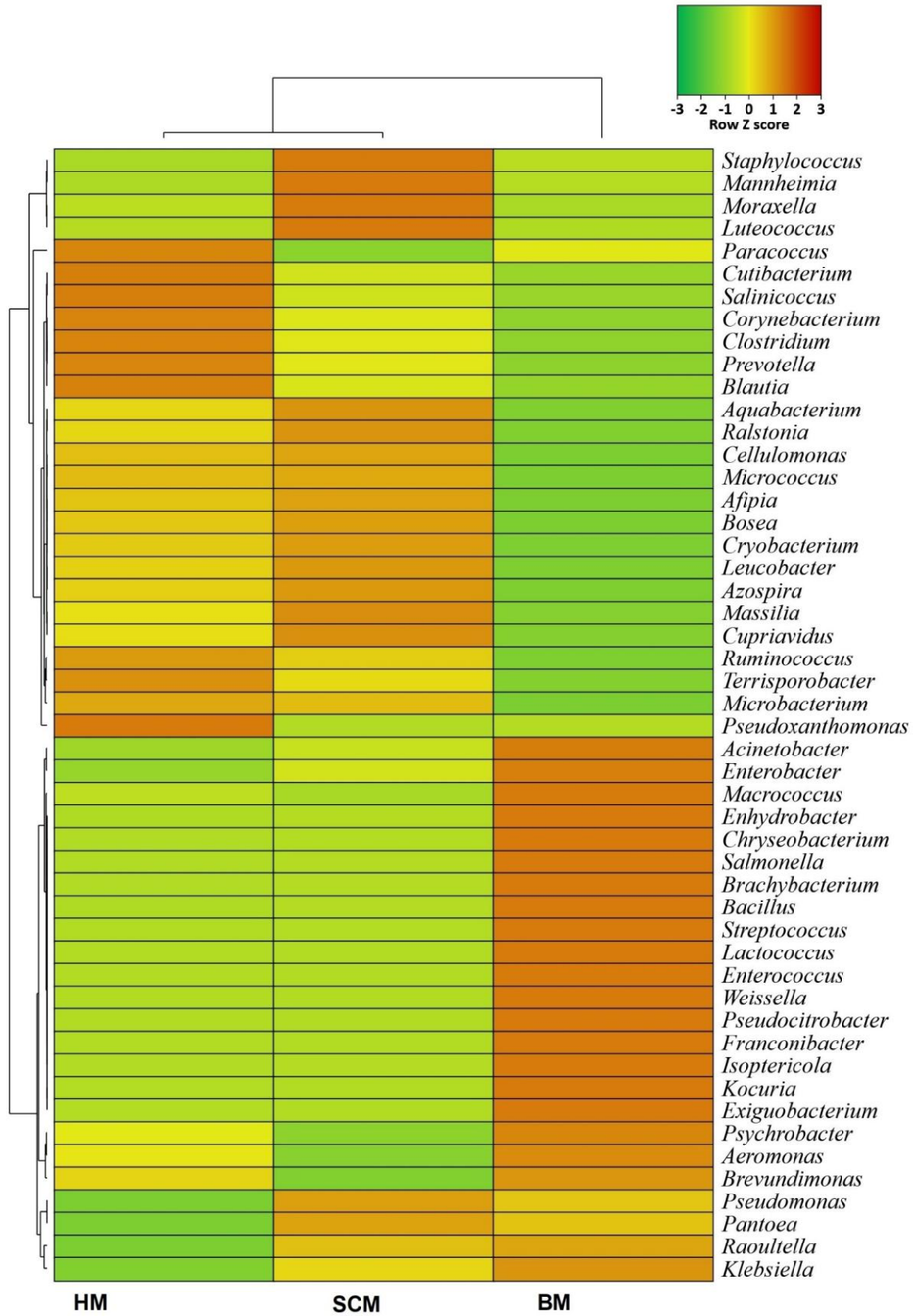


Fig 3. Taxonomic composition of microbiomes (at genus level)

Heatmap showing the composition and relative abundances of the **top abundant 50 bacterial genera** in the metagenomes of subclinical mastitis milk (SCM), bulk milk (BM) and healthy milk (HM) samples.

The less abundant orders by a given samples are shown in green color (Z score = -3) whereas those having medium abundance are presented in yellow color (Z score = 0), and orders that are more abundant represented in red color (Z score = 3).

Fe. Me. S. P. Rum.
27° international congress
Bologna, September 4-6, 2025

Results

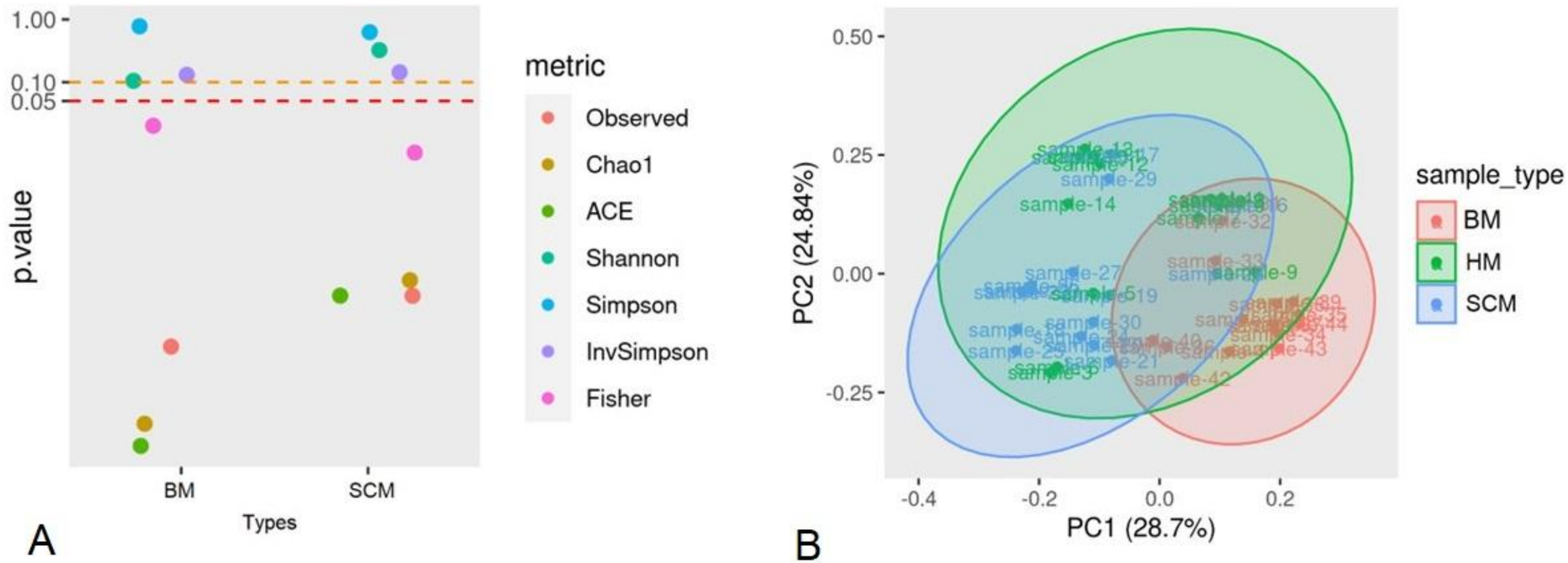
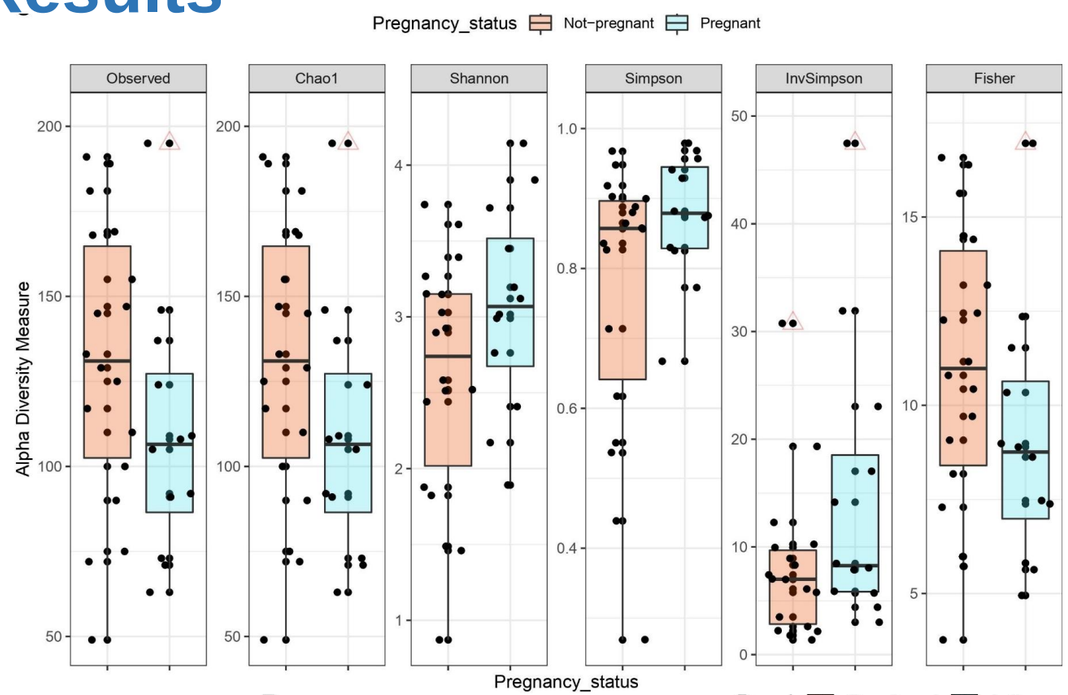


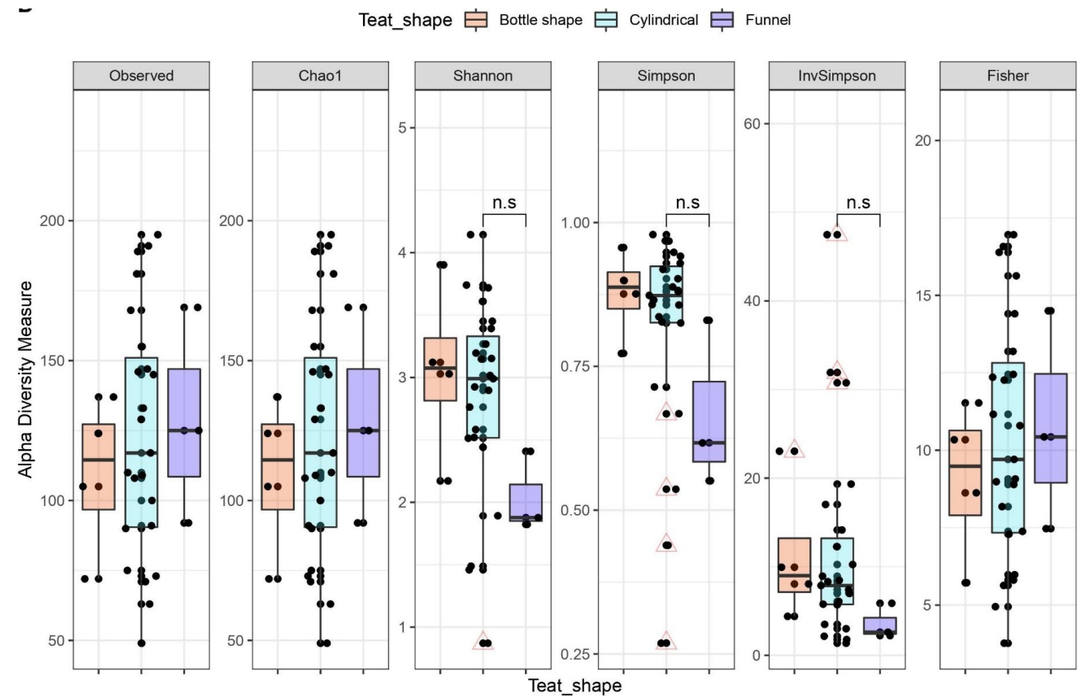
Fig 4. Microbiome diversity (Alpha & Beta diversity)

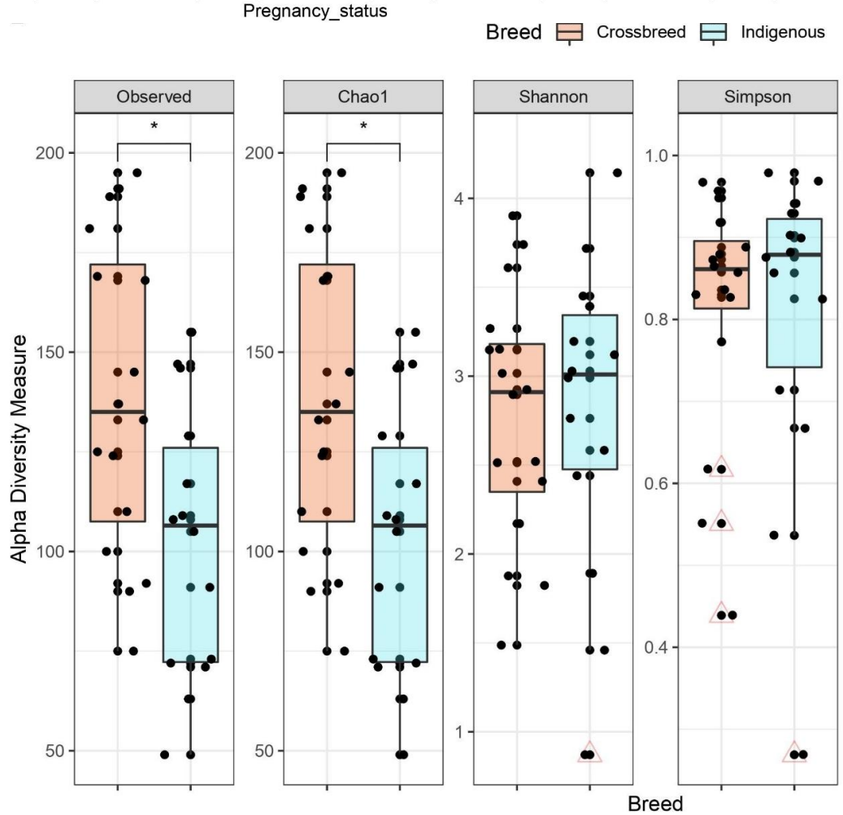
(A) Within subject **(Alpha) diversity** measure. Observed species, Chao1, ACE, Shannon, Simpson, InvSimpson and Fisher indices were estimated within sample bacterial diversity of bulk milk (BM) and subclinical mastitis milk (SCM) compared to healthy milk (HM) samples.

(B) Between-subject **(Beta) diversity** measure. Bacterial beta diversity is measured with Bray-Curtis dissi
visualized on principal coordinate analysis (PCoA) plot. The samples are colored according to subje
green: HM and blue: SCM) and joined with the respective ellipses. Pairwise comparisons on a dista
PERMANOVA test under reduced model shows significant bacterial community differences among
PERMANOVA test).

Results

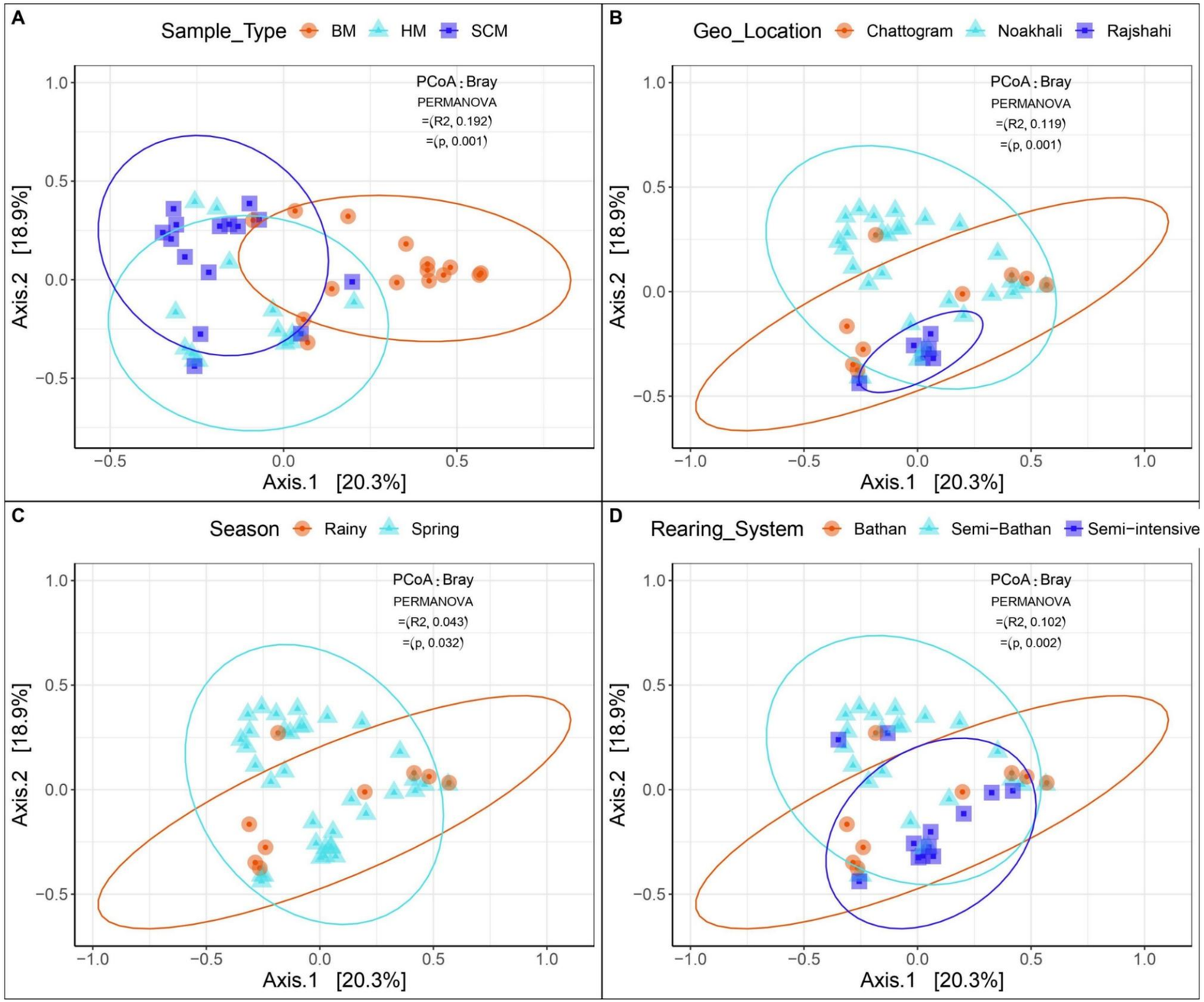






Significant differences **between the Crossbreed and the indigenous breed** in alpha diversity indices (Chao and Fisher)

Results



No different clustering of beta diversity

Fe. Me. S. P. Rum.
27° international congress
Bologna, September 4-6, 2025

Summary

These results suggest that

- Unique bacterial genera are present in different types of samples, which could help differentiate between healthy and SCM pathogens
- Indicating mammary dysbiosis and potential bacterial contamination of bulk milk in water buffalo.

Alpha-diversity analysis revealed significant differences (p = 0.005) between BM and SCM milk compared to HM.

Beta-diversity did not show distinct clusters among the sample groups.

SCM samples consistently exhibited higher bacteriome composition and abundance compared to BM and HM.

Staphylococcus dominated SCM and HM samples, whereas it was lower in BM.

BM was dominated by Staphyloco Macrococcus, Chryseobacterium and Acinetobacter.

Fe. Me. S. P. Rum.
27° international congress
Bologna, September 4-6, 2025

Thank you for your patience

ধন্যবাদ

Grazie mille

Tack

Dank u wel

Healthy Udder, Safe milk